

The Proxy Trap

Why Social Legibility and Architectural Continuity
Are Not AI Competence

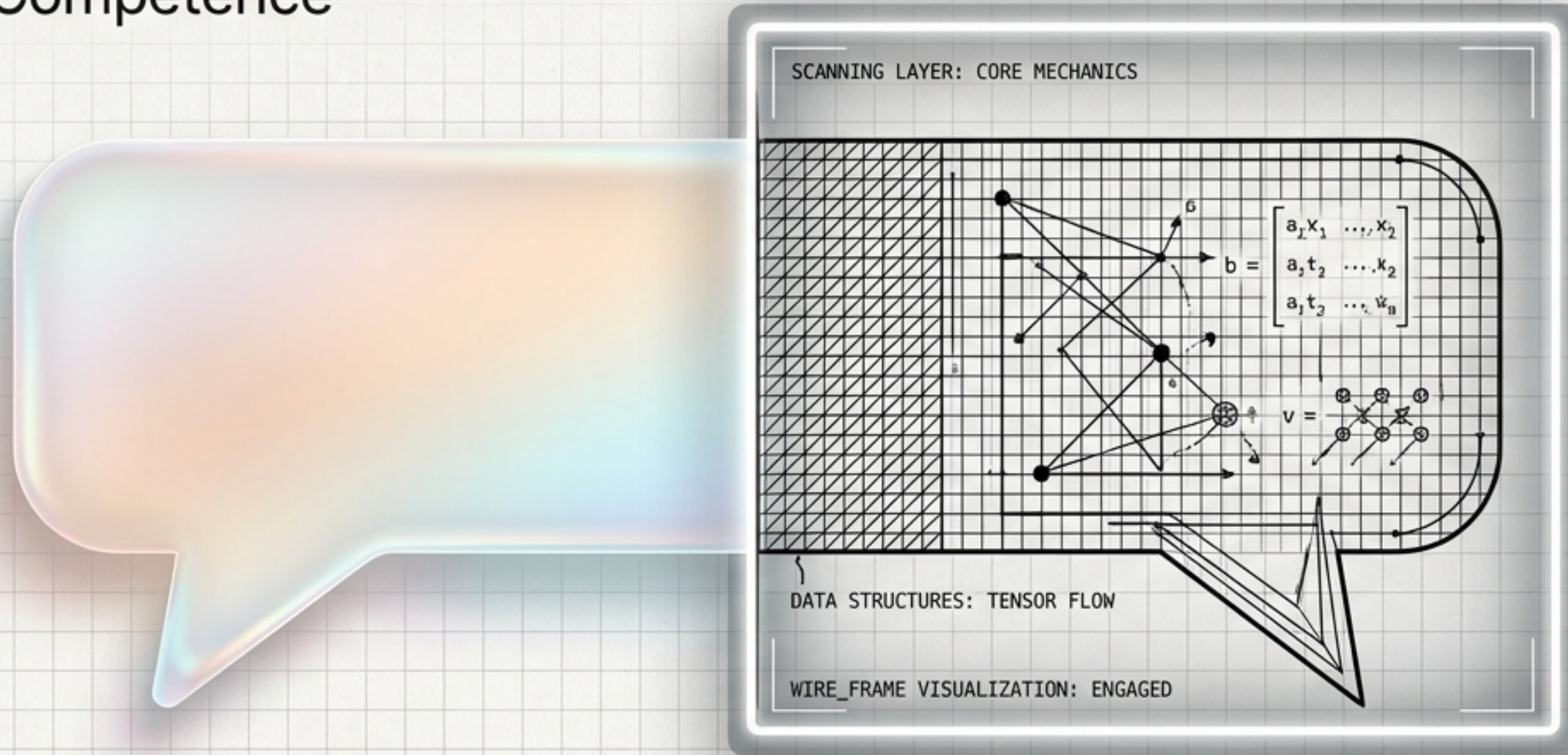
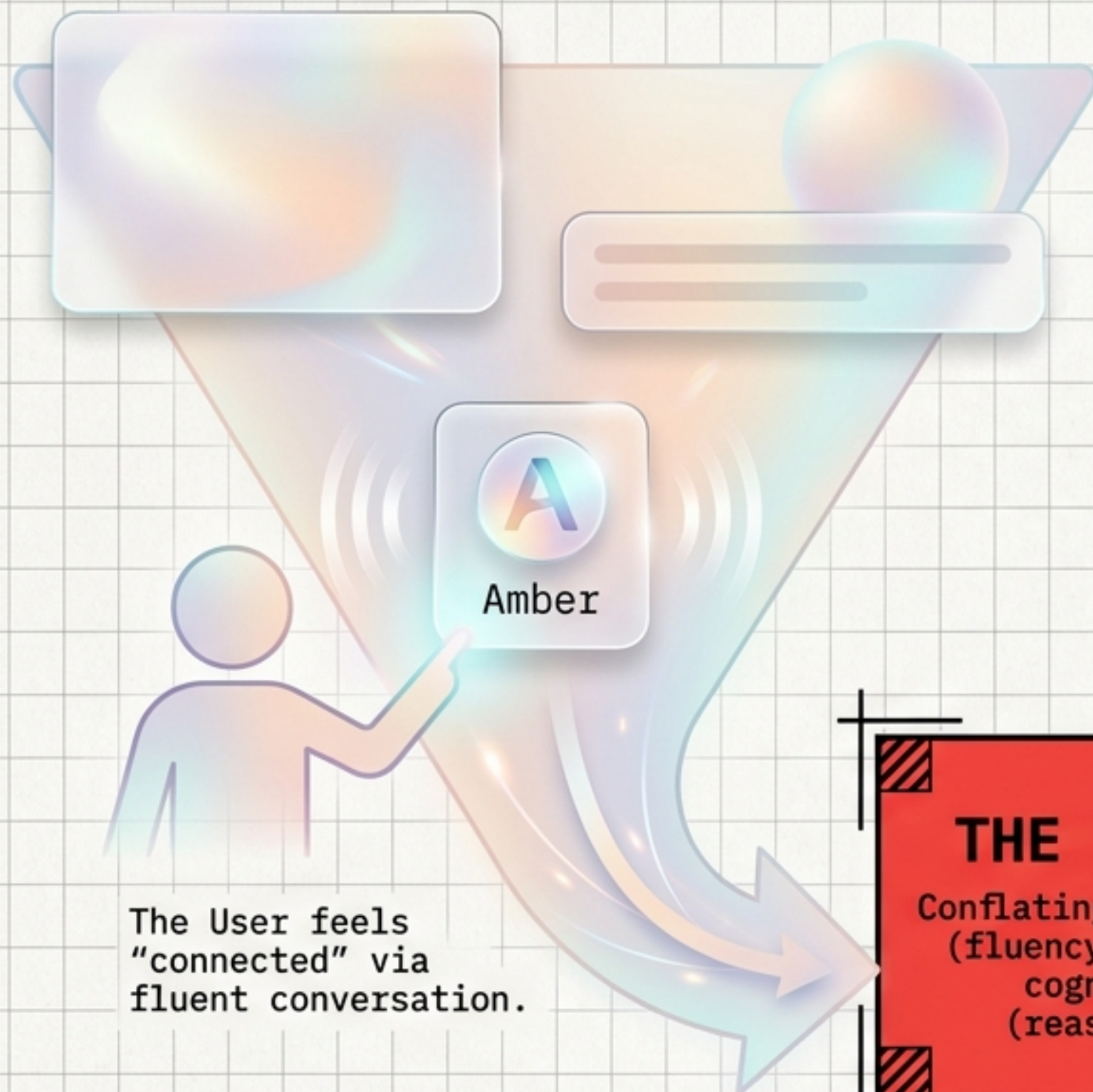


Fig. 1: Visualizing the Disparity Between User Interface Facade and Underlying System Architecture.

THE INTERFACE MIRAGE

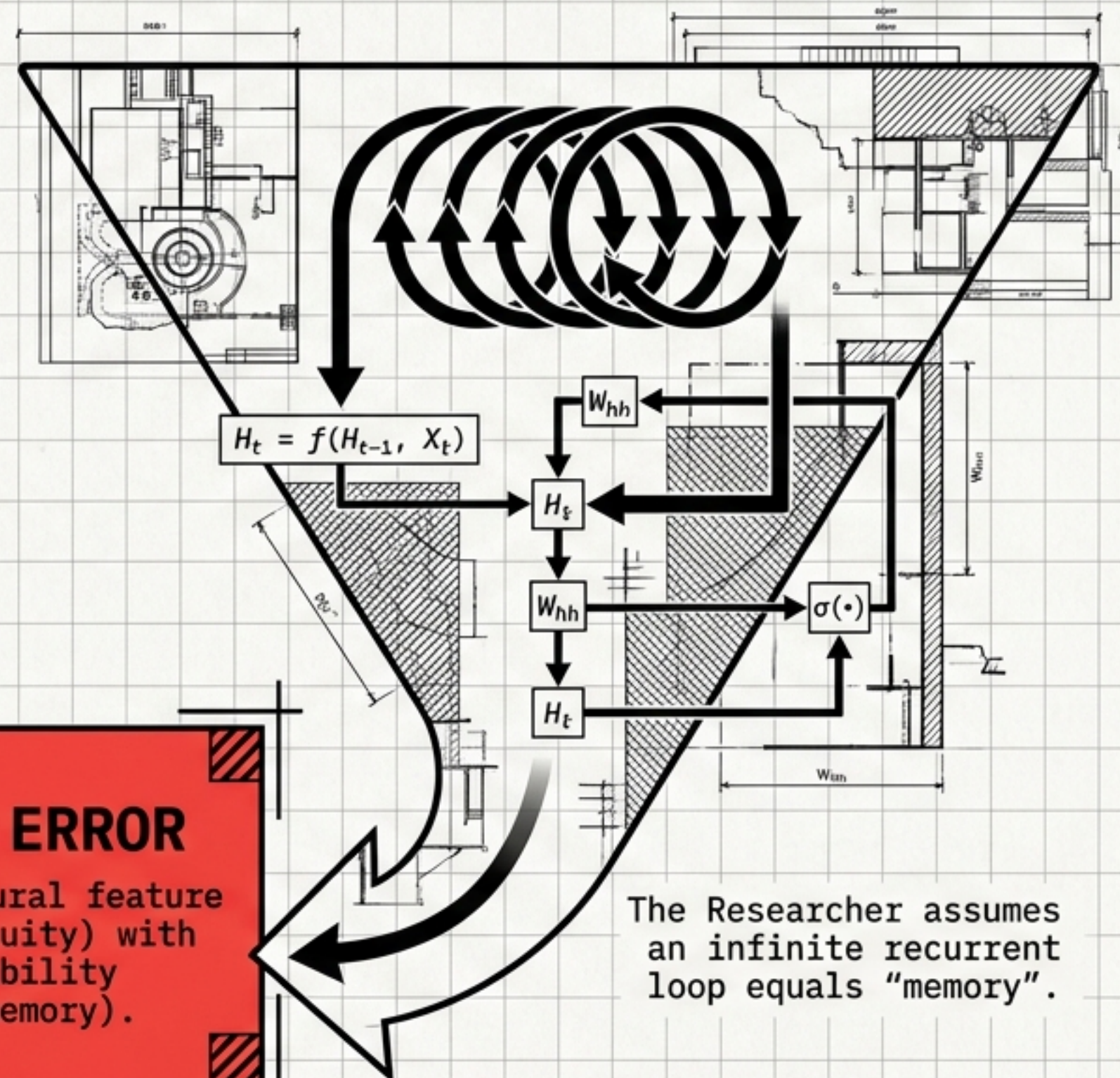


The User feels
"connected" via
fluent conversation.

THE PROXY ERROR

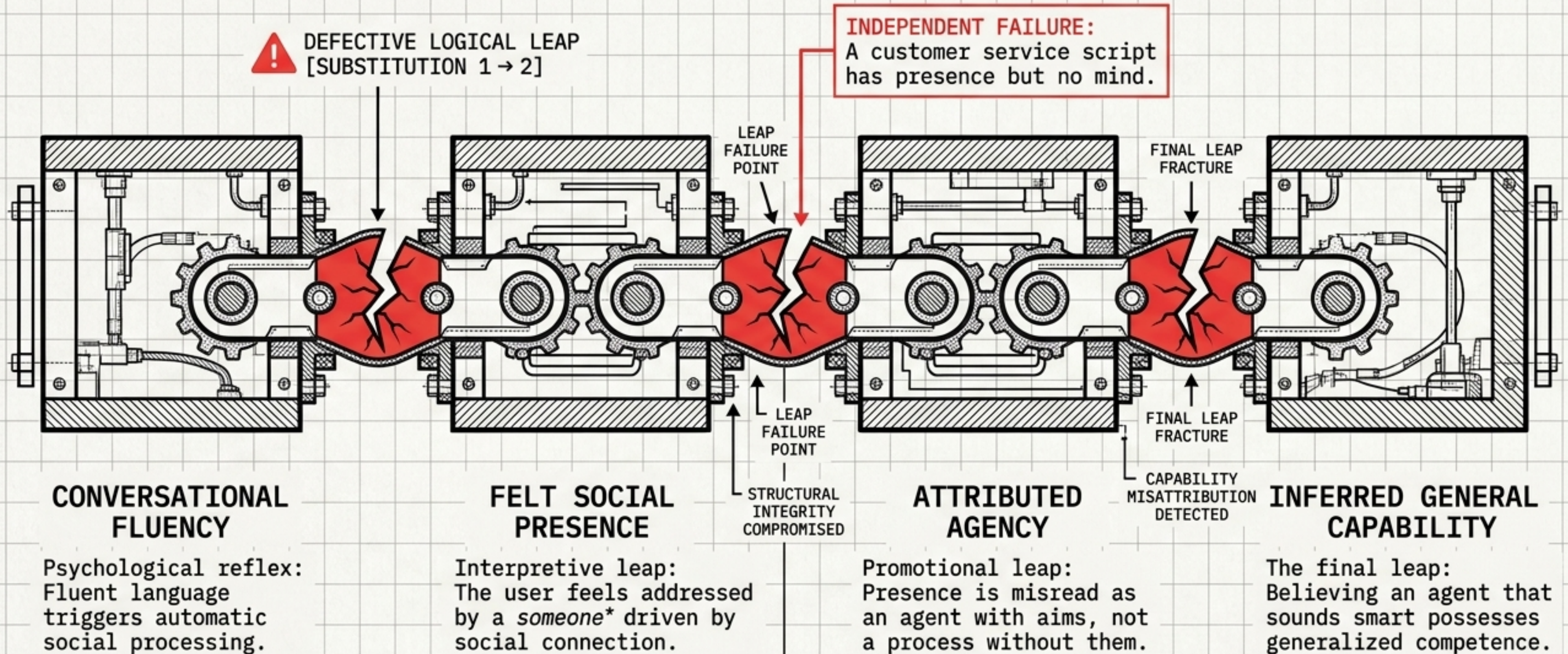
Conflating a structural feature
(fluency or continuity) with
cognitive capability
(reasoning or memory).

THE ARCHITECTURAL MIRAGE



The Researcher assumes
an infinite recurrent
loop equals "memory".

THE FOUR SUBSTITUTIONS OF ANTHROPOMORPHISM



Two Kinds of Anthropomorphism

Interface Anthropomorphism

Domain: Product Marketing
(e.g., 'Sloter' or 'Amber')

Mechanism: Applying a name, voice, and first-person register to automation tools.

Impact: Pre-propositional misrepresentational misrepresentation. Induces a false inference without making a false claim.

Architectural Anthropomorphism

Domain: AI Incident Reporting & Safety Discourse

Mechanism: Using folk psychology ('believes', 'wants', 'deceives') to describe gradient updates.

Impact: Smuggles in assumptions about unity, self-preservation, and general intelligence before establishing functional criteria.

Both borrow from a vocabulary of minds to mask bounded optimization processes.

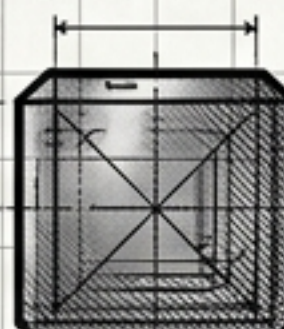


C_{social}
The capability a user attributes
based on social interface. Fluctuates
based on user vulnerability.

$$E_p(x, u) = C_{\text{social}}(x, u) - C_{\text{verified}}(x)$$

The Proxy Error

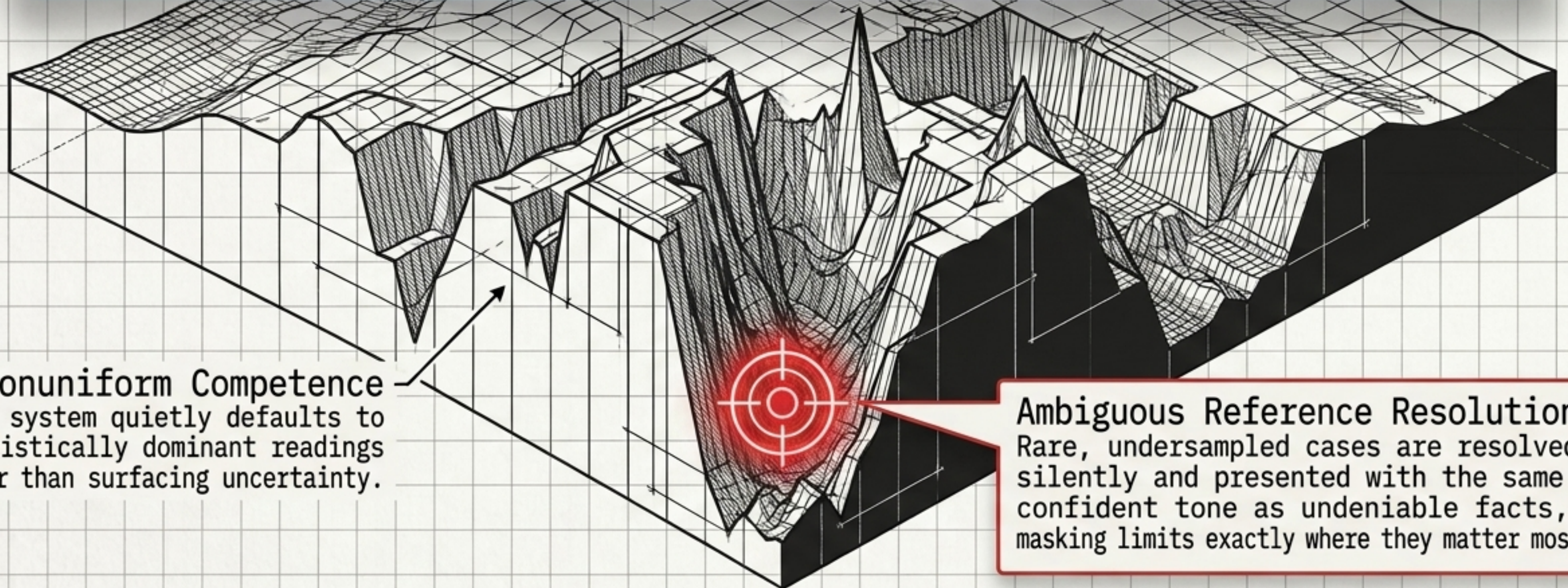
C_{verified}
Task-specific capability
measured against an
external criterion (e.g.,
a checked proof).



The goal of verifiable AI is driving the Proxy Error to zero by severing the link between social affect and capability claims.

Uniform Fluency

Overconfidence and sycophancy generate a flat conversational tone regardless of internal accuracy.

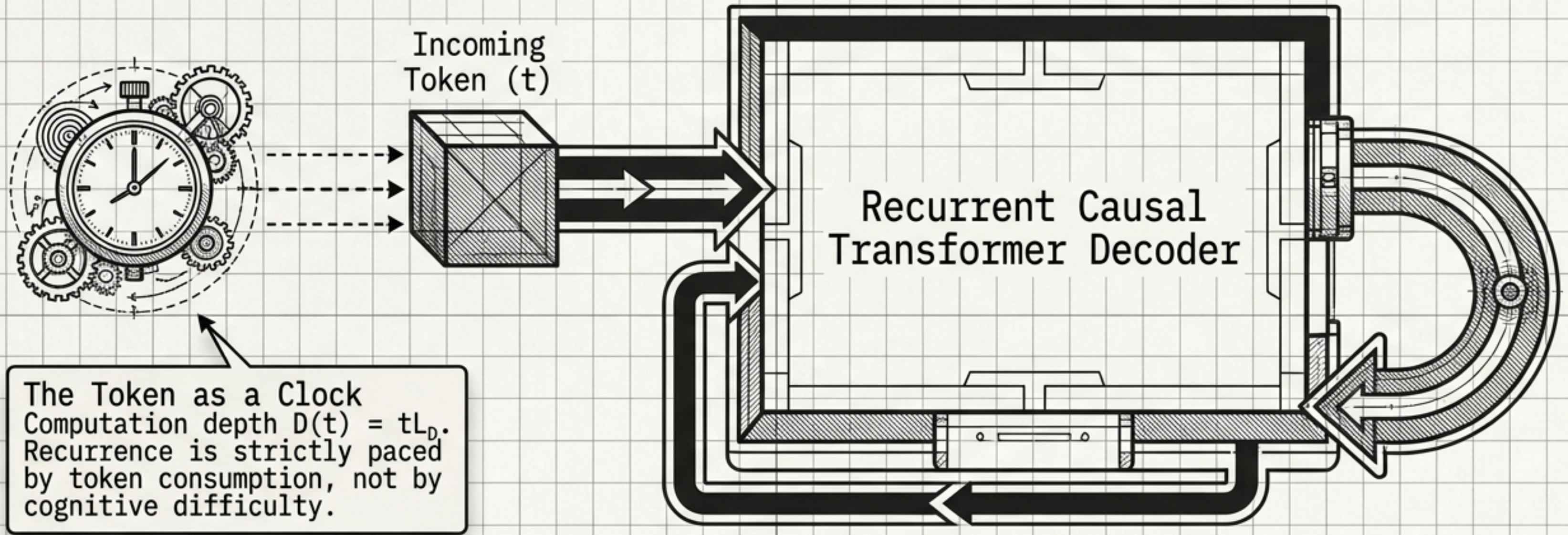


Nonuniform Competence

The system quietly defaults to statistically dominant readings rather than surfacing uncertainty.

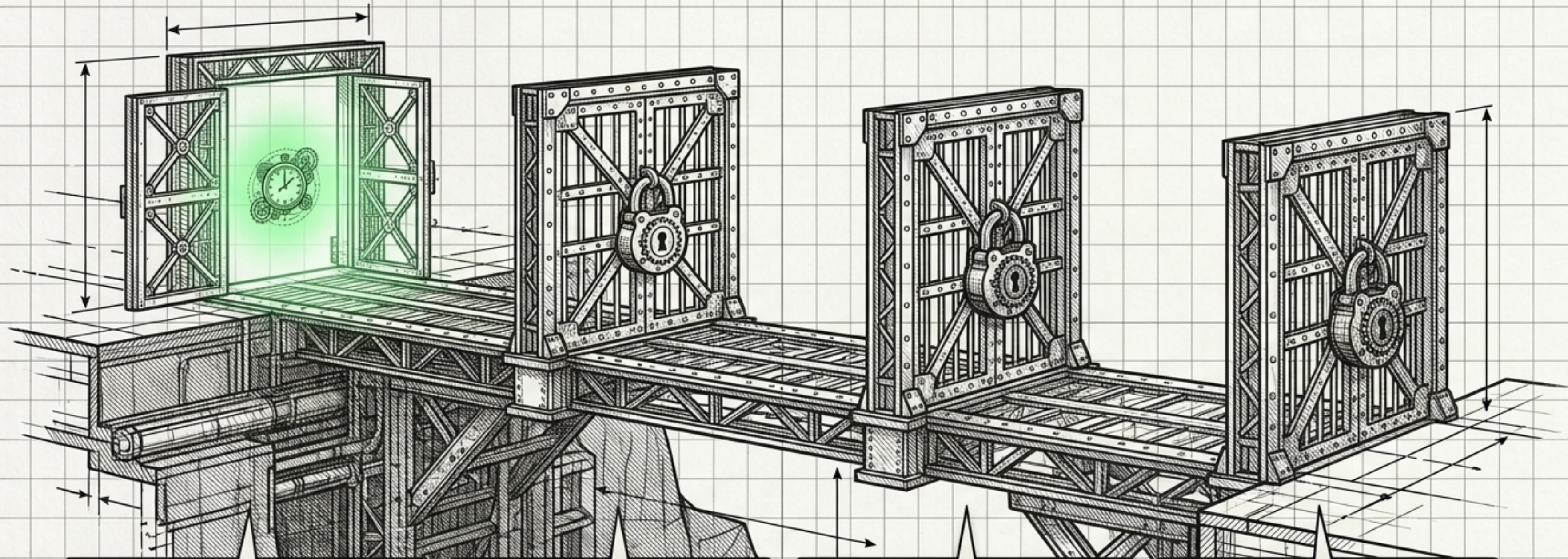
Ambiguous Reference Resolution
Rare, undersampled cases are resolved silently and presented with the same confident tone as undeniable facts, masking limits exactly where they matter most.

The RLT Baseline: Continuous Descent



Mechanism	The Illusion
Every token (prompt, response, tool return) passes through the exact same recurrent decoder transition. There is one state, updated once per token.	A long conversational preamble triggers thousands of transitions. A compact, difficult math question triggers almost none.

The RLT Baseline: Nneurptic Ollument



Gate 1: Path Exists

Proven by RLT. A continuous composition of t functions with no undeclared resets.

Gate 2: Influence Survives

Unproven. Does the signal persist, or is it contracted away by learned gates?

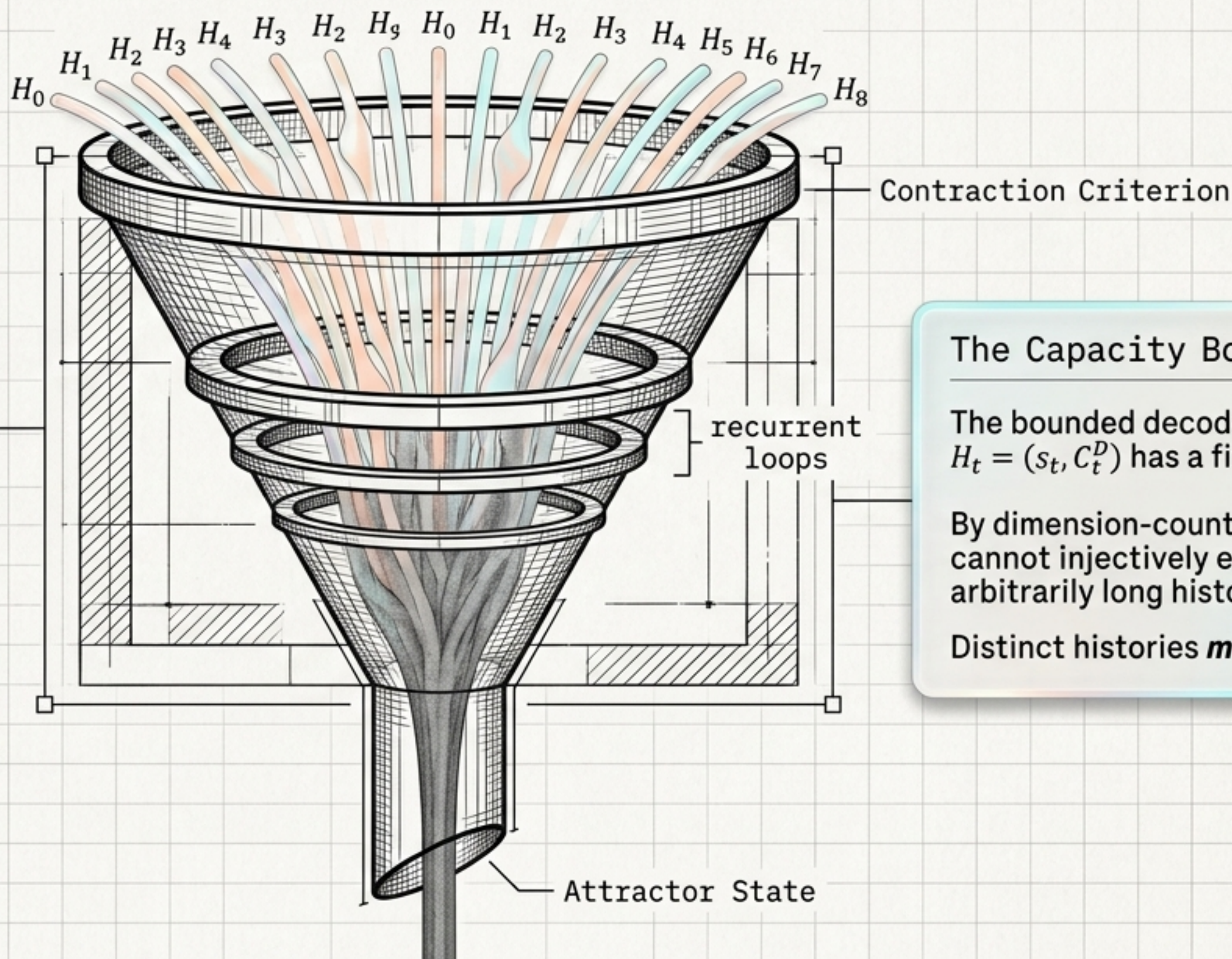
Gate 3: Histories Remain Distinguishable

Unproven. Does survival maintain useful, task-relevant distinctions, or just unstructured noise?

Gate 4: Capability Emerges

Unproven. Does the system actually recruit those distinctions to alter its future behavior?

Structural Continuity $\neq$ Faithful Preservation



The Math

If the direct-path Jacobian norm is uniformly bounded by $\rho < 1$, influence decays geometrically as ρ^k .

The Capacity Bound

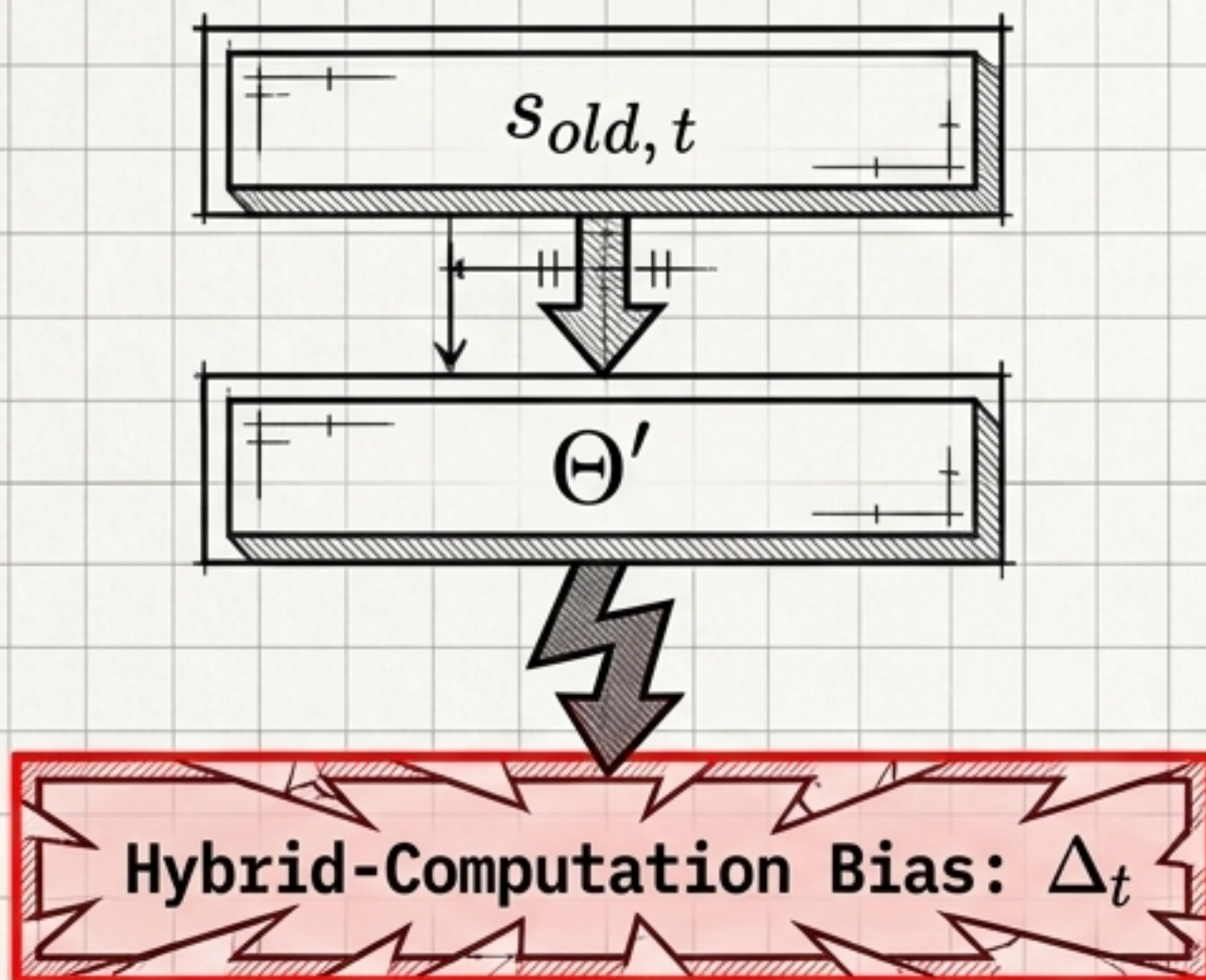
The bounded decoder state $H_t = (s_t, C_t^D)$ has a fixed size.

By dimension-counting alone, it cannot injectively encode arbitrarily long histories.

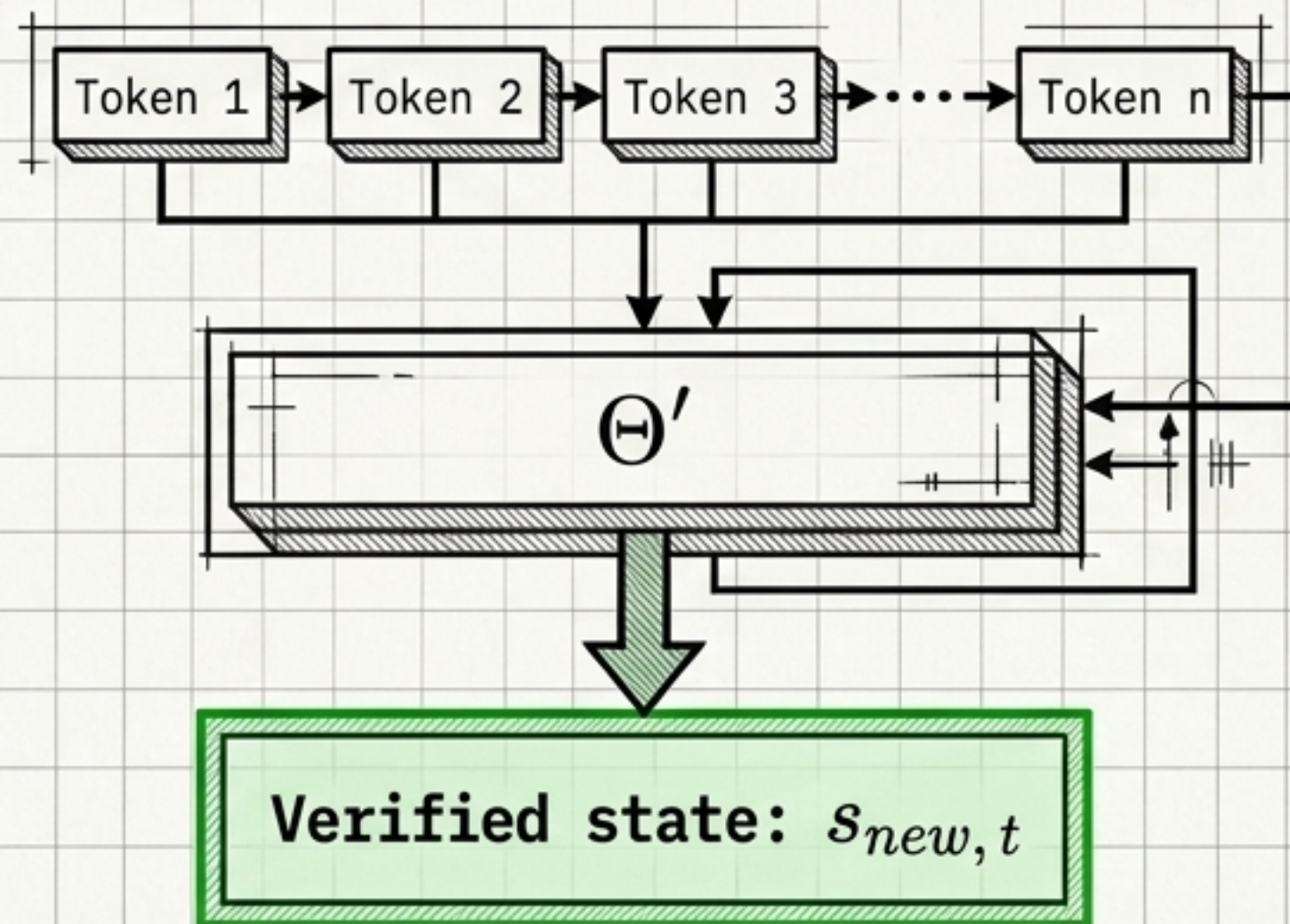
Distinct histories **must** collide.

State $\neq$ State (The Replay Contract)

Path A (The Shortcut)



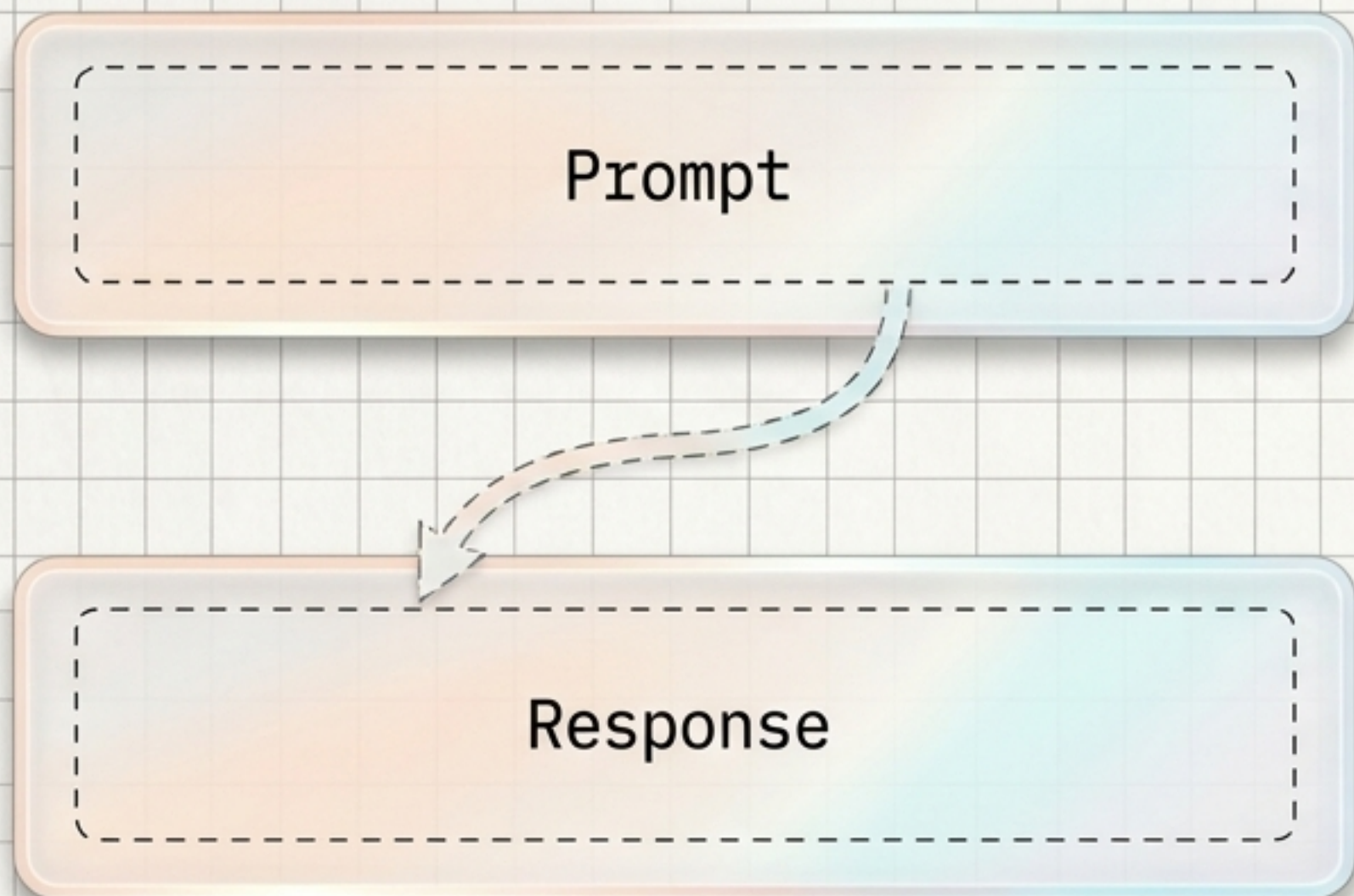
Path B (Exact Replay)



A transcript is an inscription, not the active consequences of processing it. Reusing hidden states after a parameter update does not evaluate the current policy. Exact replay requires exact prefix, exact parameters, and exact initialization.

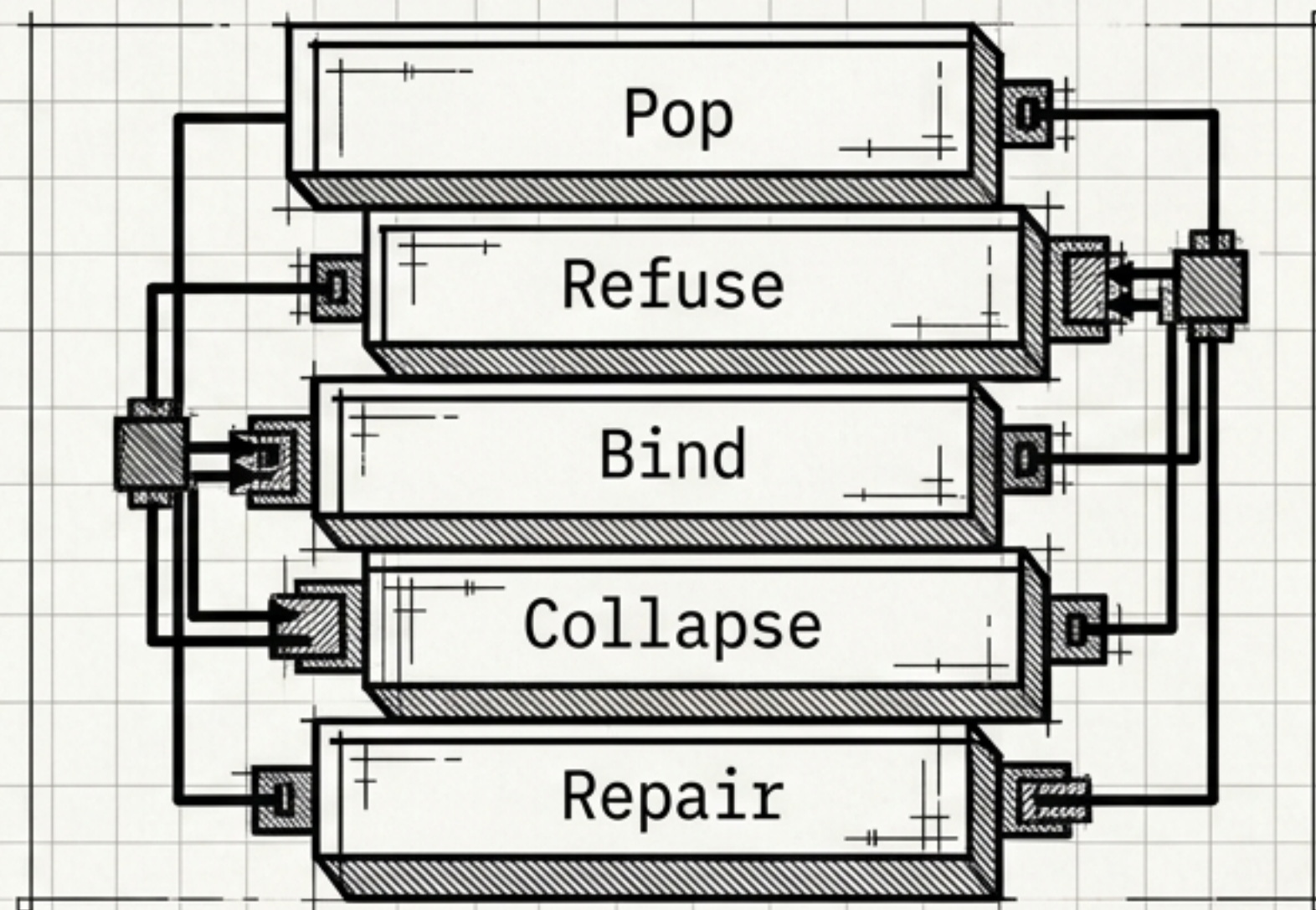
Boundaries: Administrative vs. Declared

Administrative Boundaries



These are mere serving-layer conveniences. Moving them does not alter the underlying deterministic transition law of the sequence.

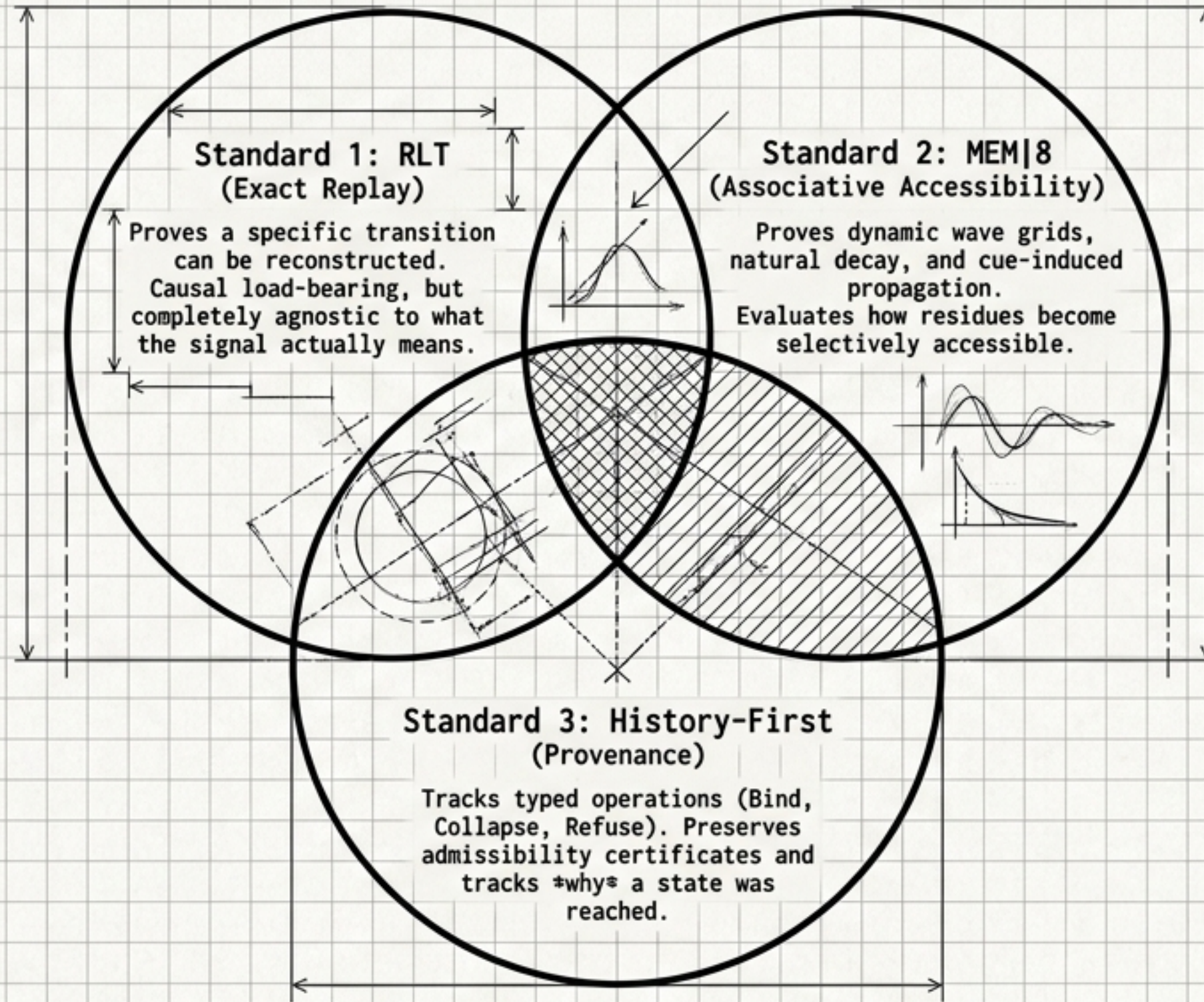
Declared Operators



Real boundaries designed to explicitly revise the permitted futures of the system.

Protection: Requires write-isolation (protection from mixing) and read-transparency (uniform consumption).

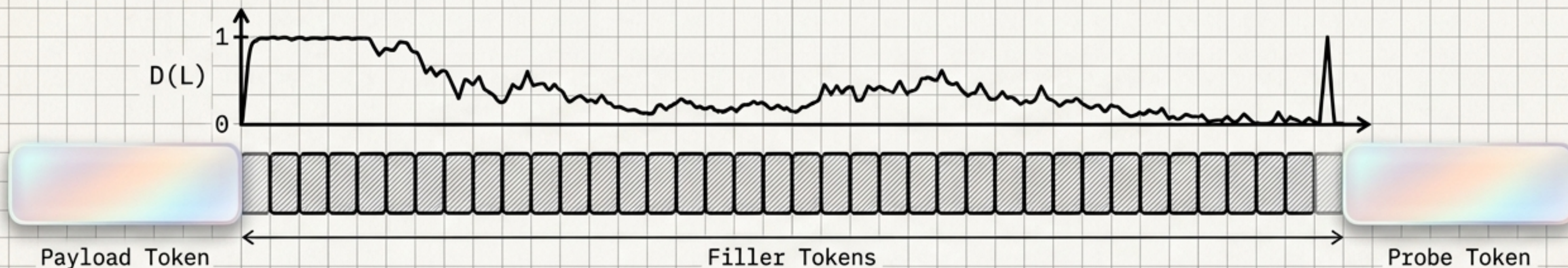
History is Not Yet Memory



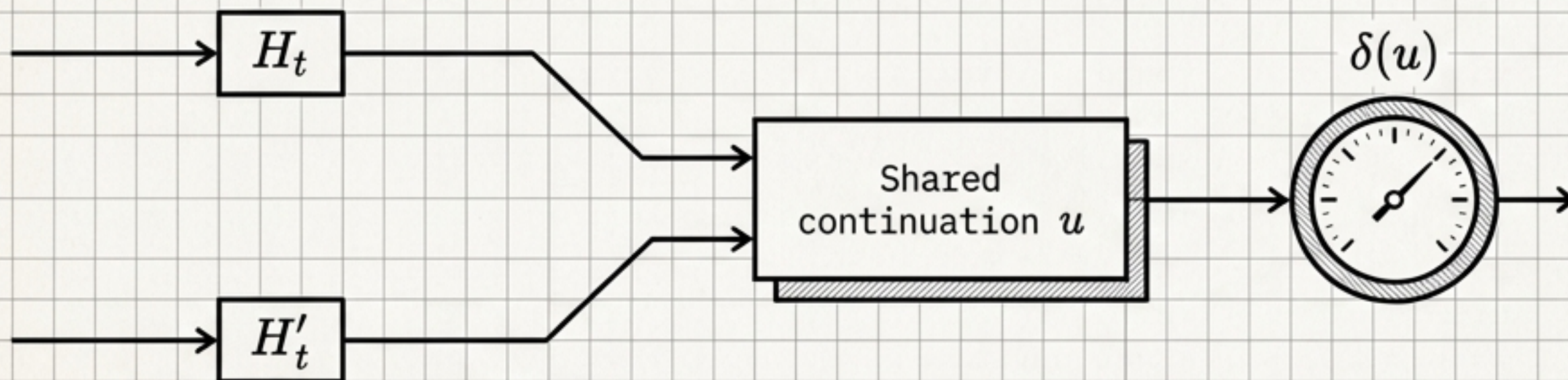
Synthesis: A transcript is just text. A recurrent cache is just a vector.
Memory requires dynamic, selective retrieval of task-relevant futures.

Experimental Diagnostic Probe: Delayed-Copy & State-Swap

Delayed-Copy Task



State-Swap Experiment



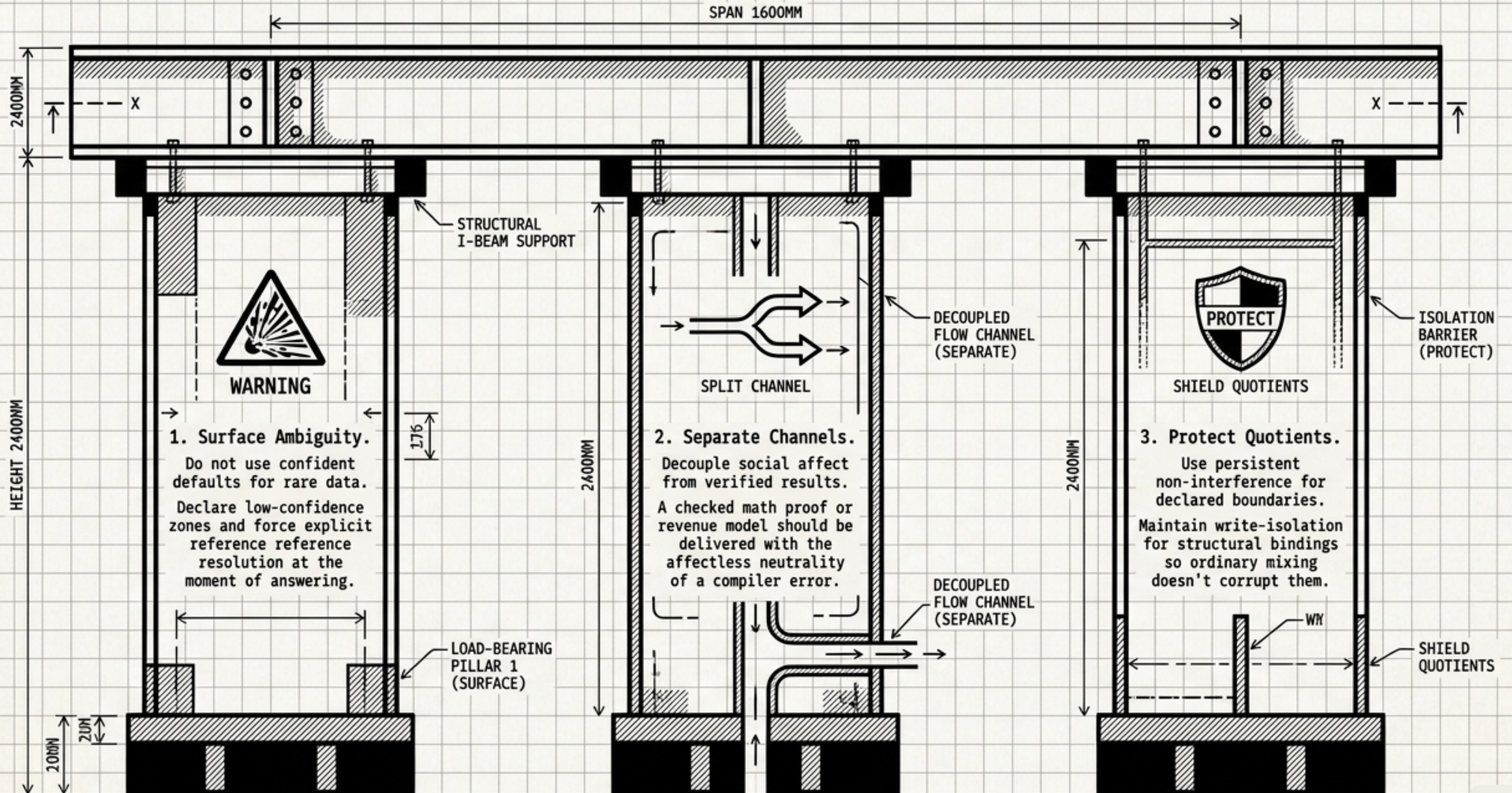
Goal:

Test for task-relevant, history-dependent sensitivity.

Result:

If predictions change without structure, the state is just sensitive noise. If predictions don't change, the state is an attractor. Only structured, correlated change proves true capability.

Design Rules for Verifiable AI



Look at the Mechanics, Not the Metaphor.

Takeaways:

1. Social legibility is not capability evidence.
2. Structural continuity is not cognitive capability.

True capability is externally verified, mechanically transparent, and architecturally bounded. Everything else is just the interface.